PRE MANUFACTURE UNIT TEST CHECKLIST

This unit test checklist will serve as a valuble tool when comissioning your PCB.

Fill this in thoroughly. Please

It will help to understand the correct function of your PCB, as well as design errors will become clear.

PCB Design Intent		
Completed:	YES	
Reviewed:	YES	

Stage	Function	Description	Verification Method	Steps	Result	Result2	Notes
Before Assembly	QUALITY CONTROL	PCB Quality Control	Inspection	Check the PCB is has been manufactured to our expected quality as well as all hardware	PASS	PASS	Back silkscreen has incorrect numbering on pins (Kicad error). Fixed with label maker
Before Assembly	CORRECTLY ASSEMBLED	All Hardware Accountable	Inspection	Check the PCB and KiCAD schemetic to ensure all components are accounted for (before assembly due to JLC assembly)	PASS	PASS	
During Assemby - Unpowered	POWER DISTRUBUTION	Power and Ground are Isolated	Test	Use a multimeter to read that the resistance between power and ground is an open loop	PASS	PASS	
During Assemby - Unpowered	Solder Header Pins	Solder Header Pins	Test	Check continuity where possible	PASS	PASS	
During Assemby - Powered	POWER INSPECTION	Regulator Quality Control	Test	Using a power supply with the pin headers, Ensuring correct connections, with the regulator, fuse and reverse polarity protection soldered, Measure the voltage ensuring 12V is only where expected, and 5V is where expected. Ensure LED's are lighting up	PASS	PASS	
During Assemby - Powered	LOGIC	LEDs	Test	Connect 12V to the input -> No LEDs should turn on (possibly CURRENT THRESHOLD)	PASS	PASS	
During Assemby - Powered	LOGIC	Pressue Test	Test	Connect 5V to PRESSURE THRESHOLD and PRESSURE OCSC pins -> PRESSURE THRESHOLD, AND PRESSURE OCSC LEDs ON	PASS	PASS	
During Assemby - Unpowered	Solder C Ref Pot	Solder C Ref Pot	Test	Verify resistance range, 0-10k ohms for pot only	PASS	PASS	Seems closer to 11.2k
Post Assembly	Turn Power On	CURRENT THRESH LED	Test	CURRENT THRESH LED Turns on	PASS	PASS	
Post Assembly	Test C Ref Pot	Check C Ref V range	Test	2.4-2.8 V	PASS	PASS	
Post Assembly	Set C Ref Pot	Set C ref to 2.572 V	Test	C Ref Pot should be a theoretical 4.152kohm	PASS	PASS	
Final Bench Test	LOGIC	CURRENT SENSE	Test	Connect 1V to CURRENT SENSE -> CURRENT THRESHOLD and CURRENT OCSC turns ON	PASS	PASS	
Final Bench Test	LOGIC	All LEDs	Demonstration	Connect power to PRESSURE THRESHOLD, and PRESSURE OCSC. Turn CURRENT SENSE voltage to 1V -> All LEDs Turn on	PASS	PASS	
Final Bench Test	LOGIC	Test Current Thresh	Test	Turn Current sense voltage to 4V -> CURRENT THRESHOLD turns OFF	PASS	PASS	
Final Bench Test	LOGIC	Pressue Thresh Test	Test	Disconnect power to PRESSURE THRESHOLD -> PRESSURE THRESHOLD and BSPD OKHS turn OFF	PASS	PASS	
Final Bench Test	LOGIC	Test Current Thresh	Test	Turn Current Sense down to 1V -> CURRENT_THRESHOLD and BSPD_OKHS turns on	PASS	PASS	
Final Bench Test	LOGIC	Test timout filter	Demonstration	Set Current Sense to 2.5V when all systems are good, disconnect PRESSURE_OCSC. Observe delay between PRRESSURE_OCSC disconnecting and BSPD OKHS going LOW -> BSPD OKHS LED turns off after 500ms delay	PASS	PASS	
Integration Test	Integration	Put on old motherboard	Demonstration	Connect onto old motherboard (possible wont fit new footprint but wires can be used), Conduct the comp test EV.7.7.4 The team must have a test to demonstrate BSPD operation at Electrical Technical Inspection. a. Power must not be sent to the Motor(s) of the vehicle during the test b. The test must prove the function of the complete BSPD in the vehicle, including the current sensor The suggested test would introduce a current by a separate wire from an external power supply simulating the Tractive System current while pressing the brake pedal	PASS		
In Service Test	WORKING	LV systems Track day with new CEN	Demonstration	Complete a successful track day with no issues	PASS		